

Research Case Studies (WiSe 2024)

Department of Computational Linguistics

Semester: Winter 2024

Submission Date: March 16th,

2025 Professor: Prof. Dr.

Achim Rettinger

**Addressing Online Harms Through Multitask Learning and Metrics Based
Evaluation**

Individual Portfolio

Author	Syed Huzaifa Ali
Field of Study	Natural Language Processing
Semester	2 nd
Matriculation Number	1734854
Email	s2syalii@uni-trier.de

Introduction and Motivation

The increasing prevalence of hate speech, toxicity, and misinformation on social media has made it necessary to have strong detection mechanisms in today's digital landscape. My work was motivated by the desire to help develop machine learning models that can identify and mitigate these issues, ensuring a safer online environment. I focused on developing and refining individual models, gathering datasets, creating proposals, and effectively presenting our findings along with documentation and reporting. The complexity of these challenges, along with the large amount of unstructured data on social platforms, inspired me to investigate various modeling approaches and explainability techniques to improve our results.

My Contributions

Throughout this research, I was actively involved in several key areas, ensuring the project was built on a strong foundation and executed effectively.

1. Proposal Creation

One of my first tasks was to create the project's initial proposal, which included stating the problem, defining the scope of the research, and identifying the main goals. I also carried out a thorough literature review to learn about the state-of-the-art in toxicity and misinformation detection so that we could properly position our work.

2. Dataset Gathering and Preprocessing

Getting high-quality data is essential to every machine learning endeavor. I was in charge of compiling pertinent datasets from multiple sources, making sure they were varied and indicative of patterns of disinformation and toxicity in the actual world.

- Collecting publicly available labeled datasets.
- Cleaning and preprocessing the data to remove inconsistencies.
- Augmenting datasets where necessary to improve model performance.

3. Defining the Research Question

I played an integral role in formulating the central research question for our study, ensuring it aligned with our objectives and the current gaps in the field.

4. Development of Individual Models

My main technological contribution was the creation of separate algorithms for the detection of disinformation and toxicity. I spent a lot of time training, optimizing, and assessing models like:

- **BERT-based models** for text classification tasks.
- **LSTM networks** to capture sequential patterns in textual data.
- **CNN-based models** to enhance feature extraction from text embeddings.

I was able to assess the performance of several architectures and determine the best configurations by experimenting with them. Our study was built on these models before moving on to more intricate, cohesive modeling techniques.

5. SHAP Analysis on Individual and Unified Models

I performed SHAP (Shapley Additive Explanations) analysis on both the individual models and the combined three-head model to make sure our models could be understood. However, we were unable to decipher the significance of these findings, thus we chose not to use these visualizing feature attributions into our presentation.

6. Final Presentation and Documentation

Findings must be communicated just as effectively as the research itself. With the assistance of my group members, I contributed in the organizing and making the final presentation, making sure that our work was understandable to both technical and non-technical audiences. I also helped with documentation efforts, which included a description of our approach, findings, and conclusions.

Reflection - Lessons Learned & Insights

This project was a transformative learning experience, offering deep insights into both the technical and practical aspects of machine learning research.

- **The importance of high-quality data:** I discovered personally that bad data cannot be made up for by even the most advanced models. Model performance is greatly impacted by data preparation and augmentation.
- **Model interpretability matters:** I became aware of how important it is to make sure our models can be explained by SHAP analysis, particularly in delicate applications like toxin detection.
- **Collaboration enhances research quality:** I was able to better understand the various viewpoints that go into enhancing a model's efficacy by working in a team.
- **Effective communication is key:** Just as crucial as carrying out the research is clearly and succinctly presenting the findings.
- **Importance of Synthetic Data:** I figured out that synthetic data can be really helpful in case where the data is limited although we should rely on synthetic data to much as that data itself is not reliable enough.

Conclusion

From creating the research concept to creating individual models, performing SHAP analysis, and presenting our results, my contributions to this project have been incredibly fulfilling. My technical machine learning skills and my capacity for teamwork in a research setting have both improved as a result of this experience. I believe my work supports ongoing efforts to create safer digital environments as toxicity and disinformation continue to pose a threat to online platforms.

**Declaration of
Authenticity** The
author(s)

Syed Huzaifa Ali

hereby declare that:

- this work is our/my own and original. As author(s) we/I are/am responsible for the provenance and veracity of all statements and claims made in this work.
- parts of this work that were taken from other works, either as quote or paraphrase, either direct or indirect, or obtained with the help of a third party are marked by respective statements of sources.
- parts that were generated or obtained using tools (like generative AI or other software) that influenced the content of this work are also marked. Such tools are transparently listed in this paper. Specifics about the usage of those tools (how they have been used and with what effect) is being reported.
- this work has not been handed in for a different academic assessment by us/me or another person.

Syed Huzaifa Ali

Signature(s)

Github Link:

https://github.com/Aastha031295/TWON_Metrics/tree/SHA